Example 7.5

Find the sum $A + B + C$ if A, B, C are given by

$$A=\begin{bmatrix}\sin^2\theta & 1\\ \cot^2\theta & 0\end{bmatrix},\ B=\begin{bmatrix}\cos^2\theta & 0\\ -\operatorname{cosec}^2\theta & 1\end{bmatrix} \text{ and } C=\begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix}.$$

Solution

By the definition of sum of matrices, we have

$$A+B+C=\begin{bmatrix}\sin^2\theta+\cos^2\theta+0 & 1+0-1\\ \cot^2\theta-\operatorname{cosec}^2\theta-1 & 0+1+0\end{bmatrix}=\begin{bmatrix}1 & 0\\ -2 & 1\end{bmatrix}.$$

Example 7.6

Determine $3B + 4C - D$ if B, C, and D are given by

$$B=\begin{bmatrix}2 & 3 & 0\\ 1 & -1 & 5\end{bmatrix},\ C=\begin{bmatrix}-1 & -2 & 3\\ -1 & 0 & 2\end{bmatrix},\ D=\begin{bmatrix}0 & 4 & -1\\ 5 & 6 & -5\end{bmatrix}.$$

Solution

$$3B+4C-D=\begin{bmatrix}6 & 9 & 0\\ 3 & -3 & 15\end{bmatrix}+\begin{bmatrix}-4 & -8 & 12\\ -4 & 0 & 8\end{bmatrix}+\begin{bmatrix}0 & -4 & 1\\ -5 & -6 & 5\end{bmatrix}=\begin{bmatrix}2 & -3 & 13\\ -6 & -9 & 28\end{bmatrix}.$$

Example 7.7

Simplify :

$$\sec\theta\begin{bmatrix}\sec\theta & \tan\theta\\ \tan\theta & \sec\theta\end{bmatrix}-\tan\theta\begin{bmatrix}\tan\theta & \sec\theta\\ \sec\theta & \tan\theta\end{bmatrix}.$$

Solution

If we denote the given expression by A, then using the scalar multiplication rule, we get

$$A=\begin{bmatrix}\sec^2\theta & \sec\theta\tan\theta\\ \sec\theta\tan\theta & \sec^2\theta\end{bmatrix}-\begin{bmatrix}\tan^2\theta & \tan\theta\sec\theta\\ \sec\theta\tan\theta & \tan^2\theta\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}.$$

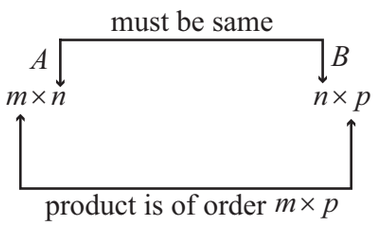
(3) Multiplication of matrices

Definition 7.15

A matrix A is said to be **conformable for multiplication** with a matrix B if the number of columns of A is equal to the number of rows of B.

That is, if $A=[a_{ij}]_{m\times n}$ and $B=[b_{ij}]_{n\times p}$ are given two matrices, then the product of matrices A and B is denoted by AB and its order is $m \times p$.

The order of AB is $m\times p=\left(\text{number of rows of } A\right)\times(\text{number of columns of } B)$.



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